

Viète's Formula: A Geometric Proof

via Inscribed Polygons and Telescoping Products

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1 Introduction

In 1593, François Viète published the remarkable identity

$$\frac{2}{\pi} = \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}} \cdots \quad (1)$$

This is the first known explicit formula for π as an infinite product, and it predates the rigorous theory of infinite products by more than a century.

The key idea is to interpret each factor in (1) as the *ratio of consecutive perimeters* of regular polygons inscribed in a unit circle, obtained by repeatedly doubling the number of sides. The product then telescopes, leaving only the ratio of the crudest possible approximation—a digon—to the circle itself.

The proof falls into two parts. The geometric heart—deriving the apothem recurrence and the perimeter ratio—rests entirely on the Pythagorean theorem, applied three

times to explicit right triangles in a unit circle. The analytic foundation—establishing that the polygon perimeters converge to the circumference—rests on the monotone convergence theorem, a fundamental property of the real numbers. Once these two parts are in place, the infinite product reduces by telescoping to the ratio of two known quantities, and Viète’s formula follows immediately.

2 Setup and Definitions

Definition 1. For $n \geq 1$, let U_n denote the perimeter of a regular 2^n -sided polygon inscribed in a unit circle (radius 1). Let a_n denote its *apothem*: the perpendicular distance from the centre to the midpoint of each side.

Figure 1 shows the first three polygons in the sequence. The apothem a_n grows from 0 toward 1 as the polygon fills out the circle: when the sides pass through the centre ($n = 1$) the apothem is zero, and as the sides press against the circumference the apothem approaches the radius.

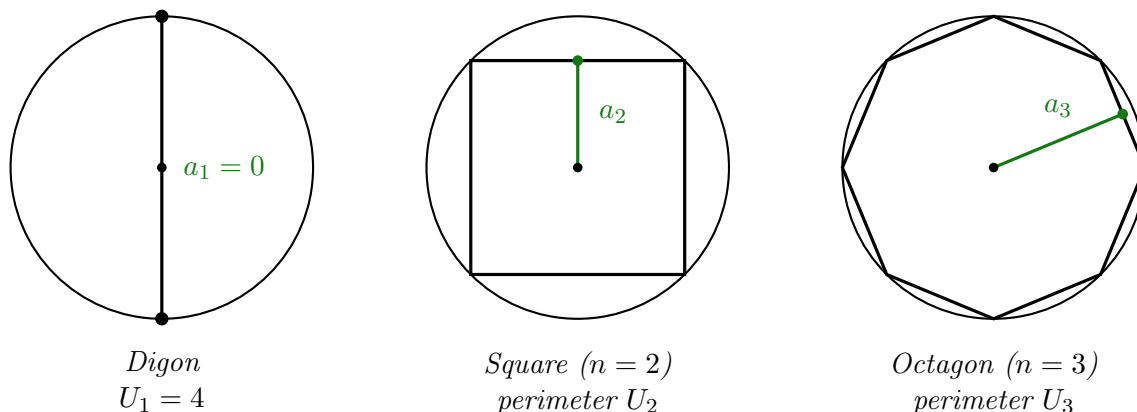


Figure 1: Regular 2^n -gons inscribed in a unit circle, with apothems (green) for $n = 1, 2, 3$. As the number of sides doubles, the polygon fills the circle more closely and the apothem approaches the radius.

2.1 The Digon: Starting Value

The leftmost polygon in Figure 1 is the *digon*: a $2^1 = 2$ -sided polygon with vertices at opposite ends of a diameter, its two sides each traversing a full diameter. Its perimeter is

$$U_1 = 2 \times 2 = 4,$$

and since its sides pass directly through the centre, the apothem is zero:

$$a_1 = 0.$$

Remark. Although geometrically degenerate (the two “sides” coincide with the same diameter), the digon is the mathematically natural base case that makes the telescoping product work cleanly.

2.2 The Sequence (U_n) Converges: Defining π

The figure suggests the definition of π directly. As the polygon fills the circle, the perimeters U_n increase toward the circumference. We take this limit as our definition of π , establishing its existence rigorously in two steps.

Step 1: (U_n) is strictly increasing. At each doubling step, a single side AB is replaced by two sides AV and VB , where V lies on the arc above AB . By the triangle inequality applied to triangle AVB :

$$|AV| + |VB| > |AB|.$$

Since every side undergoes the same replacement, the total perimeter strictly increases at every step:

$$U_1 < U_2 < U_3 < \dots$$

Step 2: (U_n) is bounded above. Consider the square whose sides are tangent to the unit circle, i.e. each side lies at perpendicular distance 1 from the centre. This square has side length 2 (each side spans a full diameter's worth of the circle) and therefore perimeter 8. Since every inscribed polygon lies strictly inside the circle, and the circle lies strictly inside the circumscribed square, every inscribed polygon lies strictly inside that square:

$$U_n < 8 \quad \text{for all } n \geq 1.$$

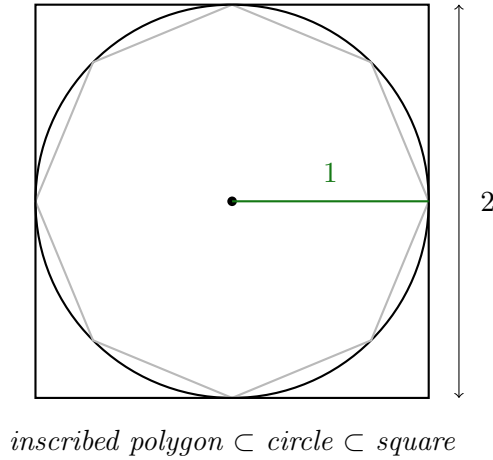


Figure 2: The containment chain: every inscribed 2^n -gon (gray) lies inside the unit circle, which lies inside the circumscribed square of side 2 and perimeter 8. This gives the upper bound $U_n < 8$.

An increasing sequence that is bounded above must converge. This is just the monotone convergence theorem (any increasing bounded sequence approaches a limit), a fundamental property of the real numbers. We therefore define:

Definition 2.

$$\pi := \frac{1}{2} \lim_{n \rightarrow \infty} U_n.$$

In other words, 2π is the limiting perimeter of the inscribed polygon sequence, which is precisely the circumference of the unit circle. We write $U_\infty = 2\pi$ as shorthand for this limit.

Remark (Consistency with the classical definition). The classical definition of π is the ratio of a circle's circumference to its diameter. We verify that Definition 2 agrees with this by a squeeze argument using circumscribed polygons in Appendix A.

3 The Core Construction

All the geometry rests on a single construction: what happens when we replace one side of the 2^{n-1} -gon with two sides of the 2^n -gon.

Take a side AB of the 2^{n-1} -gon. Let M be the midpoint of AB . Place a new vertex V at the midpoint of arc AB —the point on the unit circle directly above M . Replacing AB with the two new sides AV and VB produces the finer polygon.

The right angle at M is not assumed: it is *forced* by the circle's symmetry. The line from the centre O to the midpoint of arc AB is an axis of symmetry of the chord, and therefore meets the chord perpendicularly. This gives us three right triangles at once:

- $\triangle OAM$, right-angled at M : yields $|AM|$ from a_{n-1} ;
- $\triangle AVM$, right-angled at M : yields $|AV|$ from $|AM|$ and $|MV|$;
- the triangle formed by O , the midpoint of AV , and either endpoint of AV , right-angled at the midpoint: yields a_n from $|AV|$.

All the algebra in the proofs below is just reading lengths off these three triangles.

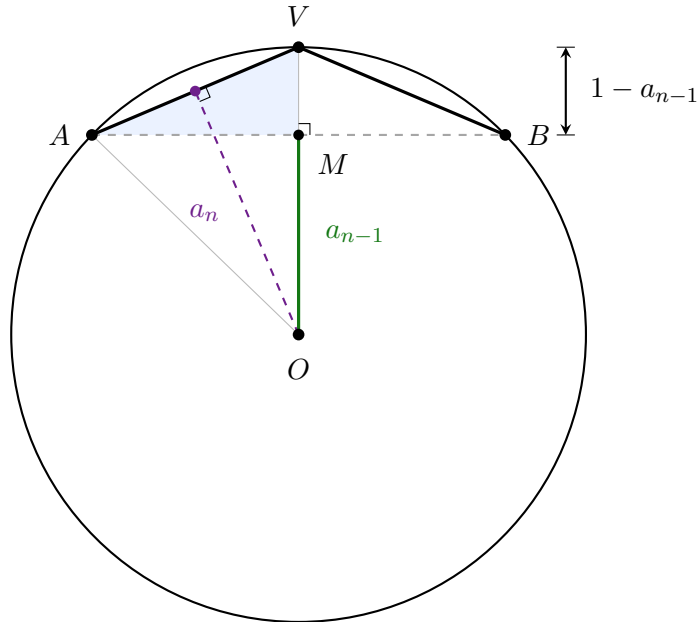


Figure 3: Arc bisection: replacing side AB of the 2^{n-1} -gon with two sides AV , VB of the 2^n -gon. The right angle at M is forced by symmetry. All quantities needed for the proof are read off the three right triangles OAM , AVM , and the triangle with hypotenuse AV .

4 The Two Lemmas

Lemma 1 (Apothem Recurrence). *The sequence (a_n) is defined inductively by*

$$a_1 = 0, \quad a_n = \sqrt{\frac{1 + a_{n-1}}{2}} \quad \text{for } n \geq 2.$$

Each term is well-defined and lies in $[0, 1)$: $a_1 = 0$, and if $a_{n-1} \in [0, 1)$ then $(1 + a_{n-1})/2 \in [1/2, 1)$, so $a_n \in [1/\sqrt{2}, 1) \subset [0, 1)$.

Proof. We apply the Pythagorean theorem to each of the three right triangles in Figure 3 in turn.

Step 1: find $|AM|$. Triangle OAM is right-angled at M , with hypotenuse $|OA| = 1$ (a radius) and one leg $|OM| = a_{n-1}$ (the old apothem). By Pythagoras:

$$|AM|^2 = 1 - a_{n-1}^2.$$

Step 2: find $|AV|$. Since $|OV| = 1$ and $|OM| = a_{n-1}$, the vertical gap between the chord's midpoint and the arc is $|MV| = 1 - a_{n-1}$. Triangle AVM is right-angled at M , with legs $|AM|$ and $|MV|$. By Pythagoras:

$$\begin{aligned} |AV|^2 &= |AM|^2 + |MV|^2 \\ &= (1 - a_{n-1}^2) + (1 - a_{n-1})^2 \\ &= (1 - a_{n-1})[(1 + a_{n-1}) + (1 - a_{n-1})] \\ &= 2(1 - a_{n-1}). \end{aligned}$$

Step 3: find a_n . The apothem a_n is the distance from O to the midpoint of the new side AV . For *any* chord of the unit circle, the centre-to-midpoint segment is perpendicular to the chord (by symmetry), giving:

$$a_n^2 + \left(\frac{|AV|}{2}\right)^2 = 1.$$

Substituting $|AV|^2 = 2(1 - a_{n-1})$:

$$a_n^2 = 1 - \frac{|AV|^2}{4} = 1 - \frac{1 - a_{n-1}}{2} = \frac{1 + a_{n-1}}{2}.$$

Taking the positive square root completes the proof. □

Remark. Since $a_n = \cos(\pi/2^n)$, the recurrence $a_n = \sqrt{(1 + a_{n-1})/2}$ is the trigonometric half-angle formula $\cos(\theta/2) = \sqrt{(1 + \cos \theta)/2}$ evaluated at $\theta = \pi/2^{n-1}$. The proof above establishes this identity for angles of the form $\pi/2^n$ from pure Euclidean geometry.

Lemma 2 (Perimeter Ratio). *For $n \geq 2$,*

$$\frac{U_{n-1}}{U_n} = a_n.$$

Proof. The 2^{n-1} -gon has 2^{n-1} sides each of length $|AB| = 2|AM|$:

$$U_{n-1} = 2^{n-1} \cdot 2|AM|.$$

The 2^n -gon has 2^n sides each of length $|AV|$:

$$U_n = 2^n \cdot |AV|.$$

Dividing:

$$\frac{U_{n-1}}{U_n} = \frac{2^{n-1} \cdot 2|AM|}{2^n \cdot |AV|} = \frac{|AM|}{|AV|}.$$

Using the expressions from Lemma 1:

$$\left(\frac{|AM|}{|AV|}\right)^2 = \frac{1 - a_{n-1}^2}{2(1 - a_{n-1})} = \frac{(1 - a_{n-1})(1 + a_{n-1})}{2(1 - a_{n-1})} = \frac{1 + a_{n-1}}{2} = a_n^2.$$

Taking the positive square root gives $U_{n-1}/U_n = a_n$. □

Remark. The apothem a_n measures how closely the 2^n -gon's flat sides approach the circle (the closer a_n is to 1, the better the fit). Lemma 2 says this *same quantity* determines how much the perimeter improves at each doubling step. The geometry and the perimeter ratio are measuring the same thing.

5 The Telescoping Product and Viète's Formula

Theorem 1 (Viète, 1593).

$$\frac{2}{\pi} = \prod_{n=2}^{\infty} a_n,$$

where $a_1 = 0$ and $a_n = \sqrt{(1 + a_{n-1})/2}$ for $n \geq 2$.

Proof. By Lemma 2, $a_n = U_{n-1}/U_n$ for each $n \geq 2$. We claim that for all $N \geq 2$,

$$\prod_{n=2}^N a_n = \frac{U_1}{U_N}.$$

The proof is by induction on N . The base case $N = 2$ gives $a_2 = U_1/U_2$, which is exactly Lemma 2 with $n = 2$. For the inductive step, assuming the identity holds for $N - 1$:

$$\prod_{n=2}^N a_n = \left(\prod_{n=2}^{N-1} a_n\right) \cdot a_N = \frac{U_1}{U_{N-1}} \cdot \frac{U_{N-1}}{U_N} = \frac{U_1}{U_N}.$$

The factor U_{N-1} cancels, and the identity is established for all $N \geq 2$. Taking the limit $N \rightarrow \infty$ and applying Definition 2:

$$\prod_{n=2}^{\infty} a_n = \lim_{N \rightarrow \infty} \frac{U_1}{U_N} = \frac{U_1}{2\pi} = \frac{4}{2\pi} = \frac{2}{\pi}. \quad \square$$

6 Expanding the Product

Applying the recurrence $a_n = \sqrt{(1 + a_{n-1})/2}$ starting from $a_1 = 0$:

$$a_2 = \sqrt{\frac{1}{2}},$$

$$a_3 = \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}},$$

$$a_4 = \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}},$$

$\vdots$

Substituting into Theorem 1:

$$\frac{2}{\pi} = \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}} \cdots$$

which is precisely Viète's formula (1). Each new factor is one more arc-bisection, one more application of Pythagoras, one more layer of nesting. The structure of the formula is the structure of the geometric construction.

7 Convergence and Numerical Verification

Define $\pi_N = 2 / \prod_{n=2}^{N+2} a_n$ as the approximation to π obtained by truncating the infinite product after its first $N + 1$ factors. By the telescoping identity of Theorem 1, $\prod_{n=2}^{N+2} a_n = U_1 / U_{N+2} = 4 / U_{N+2}$, so

$$\pi_N = \frac{U_{N+2}}{2},$$

which is simply half the perimeter of the 2^{N+2} -gon. Figure 4 shows π_N converging to π from below, as guaranteed by the strict monotonicity of (U_n) .

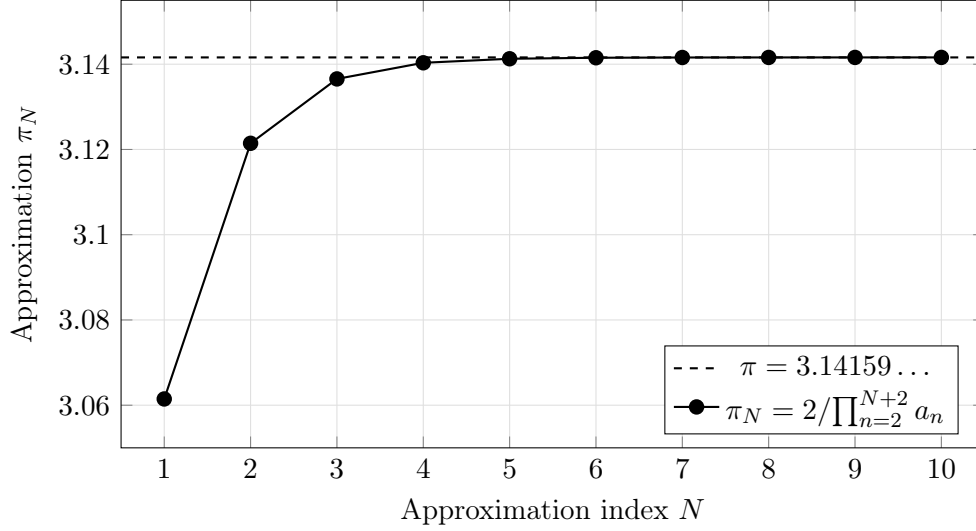


Figure 4: The partial products $\pi_N = U_{N+2}/2$ approach π from below, as guaranteed by the strict monotonicity of (U_n) . Each factor a_n is the apothem of the finer polygon; as $a_n \rightarrow 1$ the corrections become smaller and the product converges to $2/\pi$.

8 Summary

The proof rests on exactly three geometric observations, each following from a single application of Pythagoras in Figure 3.

Step	Formula	Geometric meaning
Definition	$\pi = \frac{1}{2} \lim_{n \rightarrow \infty} U_n$	(U_n) is increasing (triangle inequality) and bounded (circumscribed square); its limit defines π
Lemma 1	$a_n = \sqrt{\frac{1 + a_{n-1}}{2}}$	Arc bisection determines the new apothem via three Pythagorean steps
Lemma 2	$\frac{U_{n-1}}{U_n} = a_n$	Perimeter improvement at each doubling equals the new apothem
Theorem	$\prod_{n=2}^{\infty} a_n = \frac{U_1}{2\pi} = \frac{2}{\pi}$	Intermediate polygons telescope away; digon perimeter over circle circumference

The nested radicals in Viète's formula are a direct record of the apothems of the square, octagon, 16-gon, 32-gon, $\dots$, each computed from the last by a single arc-bisection. The infinite product telescopes because every intermediate polygon cancels, leaving only the simplest possible ratio: the perimeter of a digon divided by the circumference of the circle in which it is inscribed.

A Definition of π

This appendix verifies that our perimeter-based Definition 2 of π is consistent with the classical definition as the ratio of circumference to diameter. We sandwich the true circumference C of the unit circle between the perimeters U_n of inscribed regular 2^n -gons and V_n of the corresponding circumscribed 2^n -gons, then show that both sequences converge to the same limit.

Let V_n denote the perimeter of the regular 2^n -gon *circumscribed* about the unit circle, i.e. with sides tangent to the circle.

Lemma 3. *For each $n \geq 1$,*

$$\frac{V_n}{U_n} = \frac{1}{a_n}.$$

Proof. Consider one side AB of the inscribed 2^n -gon, with midpoint M at distance a_n from the centre O . The corresponding side of the circumscribed 2^n -gon is the tangent to the unit circle at the point T where the ray OM meets the circle; this tangent is perpendicular to OT at T and parallel to AB .

Let A' be the vertex of the circumscribed polygon adjacent to this tangent point, so that $A'T$ is half the circumscribed side, just as AM is half the inscribed side. Triangles OAM and $OA'T$ share the angle at O and are both right-angled (at M and T respectively), so they are similar. Their similarity ratio is

$$\frac{|OT|}{|OM|} = \frac{1}{a_n},$$

and therefore every corresponding half-side satisfies $|A'T| = |AM|/a_n$. Since both polygons have the same number of sides 2^n , the full perimeters satisfy

$$\frac{V_n}{U_n} = \frac{|A'T|}{|AM|} = \frac{1}{a_n}.$$

□

Squeezing to the circumference. Since every inscribed polygon lies inside the circle and every circumscribed polygon contains it, we have

$$U_n \leq C \leq V_n$$

for all n , where C denotes the circumference of the unit circle (see Appendix B for a precise justification using the supremum definition of arc length). Therefore

$$U_n \leq C \leq V_n = \frac{U_n}{a_n}.$$

We now show that the sequence (a_n) is strictly increasing and converges to 1. Base case: $a_1 = 0$. Suppose inductively that $a_{n-1} \in [0, 1)$. Then

$$0 \leq \frac{1 + a_{n-1}}{2} < 1,$$

hence $a_n = \sqrt{\frac{1+a_{n-1}}{2}} \in (0, 1)$. To prove strict monotonicity, note that for any $x \in [0, 1)$

$$\sqrt{\frac{1+x}{2}} > x.$$

Squaring (permissible since both sides are nonnegative) and rearranging yields

$$2x^2 - x - 1 < 0,$$

and the quadratic $2x^2 - x - 1$ has roots $-1/2$ and 1 , so it is negative on $(-1/2, 1)$. Thus $\sqrt{(1+x)/2} > x$ for $x \in [0, 1)$; taking $x = a_{n-1}$ gives $a_n > a_{n-1}$. Hence (a_n) is strictly increasing and bounded above by 1 , so it converges to a limit $\ell \in [0, 1]$. Passing to the limit in the recurrence $\ell = \sqrt{(1+\ell)/2}$ yields $\ell = 1$. Consequently $1/a_n \rightarrow 1$, and from the squeeze $U_n \leq C \leq U_n/a_n$ we obtain $C = \lim_{n \rightarrow \infty} U_n$. Recalling Definition 2 yields $C = 2\pi$, confirming that the above definition of π agrees with the classical ratio of circumference to diameter. $\square$

B Arc length via supremum of inscribed polygonal paths

We adopt the definition of arc length used in standard analysis texts: the length of a continuous arc $\Gamma \subset S^1$ is

$$\mathcal{L}(\Gamma) := \sup\{\ell(P) : P \text{ is a polygonal path inscribed in } \Gamma\},$$

where $\ell(P)$ denotes the Euclidean length of the polygonal path P . The circumference of the unit circle is $\mathcal{C} := \mathcal{L}(S^1)$.

Lemma 4. *For every $n \geq 1$ the perimeters U_n and V_n of the regular inscribed and circumscribed 2^n -gons satisfy*

$$U_n \leq \mathcal{C} \leq V_n.$$

Proof. The left inequality is immediate: U_n is the length of a particular polygonal path inscribed in the circle, so it cannot exceed the supremum of lengths of all such paths.

For the right inequality fix n and let the circumscribed regular 2^n -gon have consecutive tangency points $T_1, \dots, T_{2^n}$ on the circle (indices modulo 2^n) and side lengths $s_i = |T_i T_{i+1}|$. Let P be any polygonal path inscribed in the circle. By subdividing edges of P at the tangency points T_i we may assume each edge of P lies entirely within a single arc between two consecutive tangent points T_i, T_{i+1} (subdivision does not increase total length). For any two points A, B on that arc the chord length $|AB|$ is at most $|T_i T_{i+1}| = |s_i|$, because the circumscribed side s_i is a supporting segment for the convex disk and attains the maximal distance between points in the convex hull of the arc. Summing over all edges of the refined P shows $\ell(P) \leq \sum_i |s_i| = V_n$. Since P was arbitrary, taking the supremum yields $\mathcal{C} \leq V_n$. $\square$

Corollary 1. *If $V_n = U_n/\alpha_n$ with $\alpha_n \rightarrow 1$ and (U_n) is monotone increasing, then*

$$\mathcal{C} = \lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} V_n.$$

Proof. From Lemma 4 we have $U_n \leq \mathcal{C} \leq V_n = U_n/\alpha_n$ for every n . Passing to limits and using $\alpha_n \rightarrow 1$ together with monotonicity of U_n yields equality of the three limits. $\square$